实验D 列族数据库 Cassandra 实践

实验说明：

1. 实验目标

通过本实验掌握列族数据库存储架构，以及利用 Cassandra 对键空间、列族等数据进行增删改查等基本操作。

2. 实验环境

Window 系统安装 apache-cassandra-3.11.4-bin.tar.gz。

实验任务：

请完成下面题目，并将相应代码截图附于每小题后面。

(1) 创建键空间 Cus，采用简单策略，副本因子为 3。

```
(1 rows)
cqlsh> create keyspace Cus with replication = {'class': 'SimpleStrategy','replication_factor':3};
```

(2) 切换到该新建的空间并新建一个名为 pCus 的 CF（Collum Family），要求该 CF 包含：客户标识（cid）、创建时间（creattime）、客户类型(ctype)、客户性别（cgender）、职业类型(occupation)、客户姓名(cname)、身份证号（IDnum）、移动电话(tel)、住址(addr)、消费喜好(cons_pre)、累计消费金额(cumu_cons)、月消费均额(aver_cons)。

客户标识 cid 作为主键。

```
cqlsh:cus> create table pCus(
       ... cid varchar,
       ... createtime timestamp,
       ... ctype varchar,
       ... cgender varchar,
       ... occupation varchar,
       ... cname varchar,
       ... idnum varchar,
       ... tel varchar,
       ... addr varchar,
       ... cons_pre list<varchar>,
       ... cumu_cons float,
       ... aver_cons float,
       ... primary key(cid)
       ... );
cqlsh:cus>
```

(3)参考下面图片，插入 5 条模拟数据，要求数据需要覆盖下面第（4）-（9）项数据操作的需求。限定范围：客户类型两种取值（大客户 key_client，普通客户 general_client）,消费喜好取值多选（book、household goods、clothing、electronic product、camping supplies）。在此之后，查看所有数据。

```
cqlsh:cus> insert into pCus(cid,addr,aver_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('102','addB',12223.5,'male','Tom',['book', 'clothing'],dateof(now()),'gc',112345,'323456789','lawyer','13185729180');
cqlsh:cus> insert into pCus(cid,addr,ave_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('102','addB',12223.5,'male','Tom',['book', 'clothing'],dateof(now()),'gc',112345,'323456789','lawyer','13185729180');
InvalidRequest: Error from server: code=2200 [Invalid query] message="Undefined column name ave_cons"
cqlsh:cus> insert into pCus(cid,addr,aver_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('100','addA',123.5,'male','John',['book', 'clothing'],dateof(now()),'gc',12345,'123456789','teacher','13823291250');
cqlsh:cus> insert into pCus(cid,addr,aver_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('101','addC',1123.5,'female','Alice',['book', 'electronic product'],dateof(now()),'kc',22345,'223456789','doctor','18729301873');
cqlsh:cus> insert into pCus(cid,addr,aver_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('103','addE',12123.5,'male','Bob',['household goods'],dateof(now()),'kc',123345,'423456789','doctor','13698732937');
cqlsh:cus> insert into pCus(cid,addr,aver_cons,cgender,cname,cons_pre,createtime,ctype,cumu_cons,idnum,occupation,tel)
       ... values('104','addD',123.5,'female','Lisa',['book', 'household goods'],dateof(now()),'gc',2345,'523456789','teacher','13974920913');
```

```
cqlsh:cus> select * from pCus;

 cid | addr | aver_cons | cgender | cname | cons_pre                        | createtime                      | ctype | cumu_cons | idnum     | occupation | tel
-----+------+-----------+---------+-------+---------------------------------+---------------------------------+-------+-----------+-----------+------------+-------------
 104 | addD |     123.5 |  female |  Lisa |     ['book', 'household goods'] | 2023-11-30 08:36:26.198000+0000 |    gc |      2345 | 523456789 |    teacher | 13974920913
 103 | addE |   12123.5 |    male |   Bob |            ['household goods'] | 2023-11-30 08:35:28.975000+0000 |    kc | 1.2334e+05 | 423456789 |     doctor | 13698732937
 102 | addB |   12223.5 |    male |   Tom |            ['book', 'clothing'] | 2023-11-30 08:31:40.620000+0000 |    gc | 1.1234e+05 | 323456789 |     lawyer | 13185729180
 100 | addA |     123.5 |    male |  John |            ['book', 'clothing'] | 2023-11-30 08:33:12.669000+0000 |    gc |     12345 | 123456789 |    teacher | 13823291250
 101 | addC |    1123.5 |  female | Alice | ['book', 'electronic product'] | 2023-11-30 08:34:27.047000+0000 |    kc |     22345 | 223456789 |     doctor | 18729301873
```

(4)修改 cid=104 客户类型值为大客户 kc 后，查看所有数据；

```
cqlsh:cus> update pCus
       ... set ctype = 'kc'
       ... where cid = '104';
```

```
cqlsh:cus> select * from pCus;

 cid | addr | aver_cons | cgender | cname | cons_pre                        | createtime                      | ctype | cumu_cons | idnum     | occupation | tel
-----+------+-----------+---------+-------+---------------------------------+---------------------------------+-------+-----------+-----------+------------+-------------
 104 | addD |     123.5 |  female |  Lisa |     ['book', 'household goods'] | 2023-11-30 08:36:26.198000+0000 |    kc |      2345 | 523456789 |    teacher | 13974920913
 103 | addE |   12123.5 |    male |   Bob |            ['household goods'] | 2023-11-30 08:35:28.975000+0000 |    kc | 1.2334e+05 | 423456789 |     doctor | 13698732937
 102 | addB |   12223.5 |    male |   Tom |            ['book', 'clothing'] | 2023-11-30 08:31:40.620000+0000 |    gc | 1.1234e+05 | 323456789 |     lawyer | 13185729180
 100 | addA |     123.5 |    male |  John |            ['book', 'clothing'] | 2023-11-30 08:33:12.669000+0000 |    gc |     12345 | 123456789 |    teacher | 13823291250
 101 | addC |    1123.5 |  female | Alice | ['book', 'electronic product'] | 2023-11-30 08:34:27.047000+0000 |    kc |     22345 | 223456789 |     doctor | 18729301873
```

(5) occupation 创建索引后，查询所有 doctor 的信息；

```
cqlsh:cus> create index pcus_occupation_idx on pCus(occupation);
cqlsh:cus> select * from pCus
       ... where occupation = 'doctor';

 cid | addr | aver_cons | cgender | cname | cons_pre                        | createtime                      | ctype | cumu_cons | idnum     | occupation | tel
-----+------+-----------+---------+-------+---------------------------------+---------------------------------+-------+-----------+-----------+------------+-------------
 103 | addE |   12123.5 |    male |   Bob |            ['household goods'] | 2023-11-30 08:35:28.975000+0000 |    kc | 1.2334e+05 | 423456789 |     doctor | 13698732937
 101 | addC |    1123.5 |  female | Alice | ['book', 'electronic product'] | 2023-11-30 08:34:27.047000+0000 |    kc |     22345 | 223456789 |     doctor | 18729301873

(2 rows)
```

(6) ctype 未创建索引的情况下，查询所有大客户数据；

```
cqlsh:cus> select * from pCus
       ... where ctype = 'kc'
       ... allow filtering;

 cid | addr | aver_cons | cgender | cname | cons_pre                          | createtime                      | ctype | cumu_cons | idnum     | occupation | tel
-----+------+-----------+---------+-------+-----------------------------------+---------------------------------+-------+-----------+-----------+------------+-------------
 104 | addD |     123.5 |  female |  Lisa |       ['book', 'household goods'] | 2023-11-30 08:36:26.198000+0000 |    kc |      2345 | 523456789 |    teacher | 13974920913
 103 | addE |   12123.5 |    male |   Bob |           ['household goods'] | 2023-11-30 08:35:28.975000+0000 |    kc | 1.2334e+05 | 423456789 |     doctor | 13698732937
 101 | addC |    1123.5 |  female | Alice | ['book', 'electronic product'] | 2023-11-30 08:34:27.047000+0000 |    kc |     22345 | 223456789 |     doctor | 18729301873

(3 rows)
cqlsh:cus>
```

(7) cgender 未创建索引的情况下，查询所有男性客户的月均消费额度；

```
cqlsh:cus> select * from pCus
       ... where occupation = 'doctor'
       ... allow filtering;

 cid | addr | aver_cons | cgender | cname | cons_pre                          | createtime                      | ctype | cumu_cons | idnum     | occupation | tel
-----+------+-----------+---------+-------+-----------------------------------+---------------------------------+-------+-----------+-----------+------------+-------------
 103 | addE |   12123.5 |    male |   Bob |           ['household goods'] | 2023-11-30 08:35:28.975000+0000 |    kc | 1.2334e+05 | 423456789 |     doctor | 13698732937
 101 | addC |    1123.5 |  female | Alice | ['book', 'electronic product'] | 2023-11-30 08:34:27.047000+0000 |    kc |     22345 | 223456789 |     doctor | 18729301873

(2 rows)
```

(8) cgender 创建索引后，查询所有女性客户列表中的最低月均消费额；

```
cqlsh:cus> create index pcus_cgender_idx on pCus(cgender);
cqlsh:cus> select min(aver_cons) from pCus
       ... where cgender = 'female';

 system.min(aver_cons)
-----------------------
                 123.5

(1 rows)

Warnings :
Aggregation query used without partition key
```